Yung-Chin (Jim) Chen

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Research Interests

LLM Quantization; ML Systems and Accelerators; In-Memory Computing (IMC); Computer Architecture; VLSI Design

Education

Princeton University, *Ph.D. Candidate in Electrical and Computer Engineering* — Princeton, NJ
Advisor: Prof. Naveen Verma; GPA: 3.88 / 4.00 — *2024–2029 (Expected)*

National Taiwan University, *B.S. in Electrical Engineering* — Taipei, Taiwan
GPA: 3.99 / 4.00; Phi Tau Phi Honorary Member (for top 1% of college graduates) — *2019–2024*

Publications

LLM Quantization and IMC-specific Quantization

[1] **Y.-C. Chen**, C.-Y. Chen, and N. Verma. "IMC-CLINIC: Coupled Loss-Informed Newton Iterations for Clipping in Analog In-Memory Computing." *on arXiv (Under review at ICLR 2027).*

[2] **Y.-C. Chen**, O. Stouraiti, and N. Verma. "SMART-IMC: Sensitivity-Aware Mixed ADC Resolution for Transformer Inference with Analog In-Memory Computing." *on arXiv (Submitted to MLSys 2027).*

[3] **Y.-C. Chen**, C. P. Lee, Z.-W. Liou, and N. Verma. "Massive Spikes in LLMs are Bias Vectors: Mechanistic Uncovering and Spike-Free Quantization." *In Findings on EMNLP 2026.*

IMC Accelerator Architecture

[4] T.-Y. Wu*, Y.-S. Pan*, **Y.-C. Chen***, W.-C. Tseng*, et al. "N2N-BP^{2}CIM: A 28nm End-to-End Bit-Parallel, Bit-Parallel Computing-In-Memory DNN Processor with High-bandwidth 14T-SRAM Bitcell." *In TCAS-II 2026.* (* **Co-first author**)

[5] W. Zhang, S. Ando, **Y.-C. Chen**, S. Miyagi, S. Takamaeda-Yamazaki, and K. Yoshioka. "PaCiM: A Sparsity-Centric Hybrid Compute-in-Memory Architecture via Probabilistic Approximation." *In ICCAD 2024.*

[6] **Y.-C. Chen**, S. Ando, D. Fujiki, S. Takamaeda-Yamazaki, and K. Yoshioka. "HALO-CAT: A Hidden Network Processor with Activation-Localized CIM Architecture and Layer-Penetrative Tiling." *on arXiv.*

Macro-Level IMC Circuits

[7] **Y.-C. Chen**, S. Ando, D. Fujiki, S. Takamaeda-Yamazaki, and K. Yoshioka. "OSA-HCIM: On-The-Fly Saliency-Aware Hybrid SRAM CIM with Dynamic Precision Configuration." *In ASP-DAC 2024.*

[8] S. Ando, S. Miyagi, W. Zhang, **Y.-C. Chen**, and K. Yoshioka. "A 4541 TOPS/W Saliency-Aware Analog Computing In-Memory Macro with Charge-Domain Saliency Detector." *In ESSERC 2025.*

[9] W. Zhang, S. Ando, **Y.-C. Chen**, and K. Yoshioka. "ASiM: Modeling and Analyzing Inference Accuracy of SRAM-Based Analog CiM Circuits." *In TVLSI 2025.*

For a full publication list, see Google Scholar.

Research Experience

Princeton University — Princeton, NJ
Graduate Researcher; Advisor: Prof. Naveen Verma — *Jan 2025–Present*

- ***Quantization Algorithms for Analog IMC*** — Developed hardware-aware algorithms for operand clipping [1] and mixed-precision IMC computation [2] to improve LLM inference accuracy.
- ***LLM Quantization & Activation Spikes*** — Uncovered the mechanism of massive activation spikes in LLMs and leveraged their structural role to enable spike-free post-training quantization [3].

EnCharge AI Inc. — Santa Clara, CA
Research Intern; Supervisor: Chia-Yu Chen — *Aug–Dec 2026 (Expected)*

- ***Quantization*** — Developing quantization methods for efficient language and vision model inference on IMC architectures.

Keio University — Kanagawa, Japan
Researcher; Advisor: Prof. Kentaro Yoshioka — *Sep 2022–Feb 2024*

- ***IMC Circuits & Macros*** — Developed saliency-aware IMC circuit designs for convolutional neural networks [7, 8] and explored bit-parallel analog IMC input scheme (contributions mentioned in [ISSCC'24 / JSSC'25]).
- ***IMC Architectures & Modeling*** — Developed a circuit-aware IMC simulator [9], a sparsity-centric IMC architecture [5], and algorithm–architecture co-design for emerging neural networks [6].

National Taiwan University Taipei, Taiwan
Researcher; Advisor: Prof. Tsung-Te Liu *Sep 2021–Feb 2024*

- ***IMC Accelerator Chip*** — Developed and taped out a 28nm end-to-end SRAM IMC processor [4].

Honors and Awards

Study Abroad Scholarship — Japan Student Services Organization *Sep 2022–Jul 2023*
Irving T. Ho Memorial Scholarship — EE Department, National Taiwan University *Dec 2022*
The Memorial Scholarship Foundation to Lin Hsiung Chen (acceptance rate: 2%) *Nov 2022*
Research Grant — National Science and Technology Council (NSTC), Taiwan *Jul 2022–Jan 2023*
Research Grant — Taiwan Semiconductor Manufacturing Co., Ltd. (TSMC) *Feb 2022–Jun 2022*
1st Place (Best Solver Award) — MakeNTU, Taiwan's largest student maker hackathon *Mar 2022*
Principal Award — Awarded to the top 2% of NTUEE students for academic excellence *Dec 2022*

Technical Skills

Programming: Python, C++, SystemVerilog, Verilog, MATLAB
ML Libraries: PyTorch, Triton, TensorFlow, NumPy
EDA & Hardware: Design Compiler, Innovus, Virtuoso, HSPICE, NC-Verilog, FPGA
Systems & Tools: Linux, Git, Slurm, LaTeX

Teaching Experience

Teaching Assistant, Design of VLSI Systems — Prof. Naveen Verma *Sep 2025–Dec 2025*
Teaching Assistant, Computer Architecture — Prof. Tsung-Te Liu *Sep 2023–Jan 2024*
Teaching Assistant, EECS Lab Undergraduate Research — Prof. Tsung-Te Liu *Sep 2023–Jan 2024*
Teaching Assistant, Signals and Systems — Prof. Lin-Shan Lee *Feb 2022–Jun 2022*

Service

Reviewer, ICLR *2026*
Reviewer, TCAS-I *2026*